Comparing Distributions and Compound Events

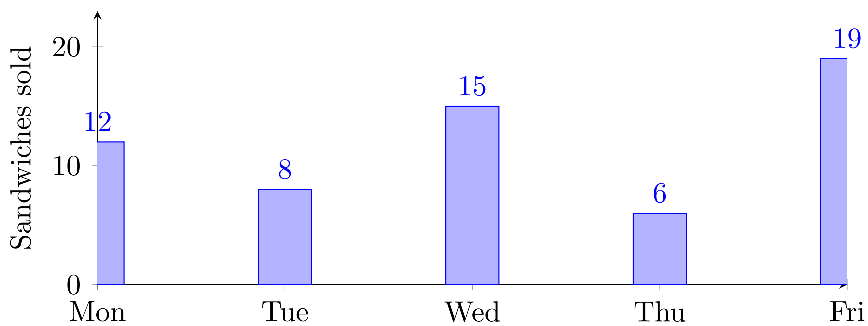
Comparing centre and spread, reading a data display, and finding compound-event probabilities

Directions

- **Centre** says where a data set sits: the mean is the fair-share value, the median is the middle value once the data are in order.
- **Spread** says how stretched it is: the range is the largest value minus the smallest. Two sets can share a centre and still look nothing alike.
- **Compound events** happen together. Count the favourable outcomes and the total outcomes, then write the probability as a fraction in lowest terms.
- Leave probabilities as exact fractions, never as decimals.

Display A — sandwiches sold at the school café last week

Problems 3, 7 and 10 all read this chart. No other problem uses it.



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1. The Falcons scored 4, 6, 7, 9, 9 points in their five games. Find the mean number of points per game.

Answer: ________________

2. The Herons scored 5, 5, 8, 10, 12 points in their five games. Find the mean number of points per game.

Answer: ________________

3. Using Display A, how many sandwiches were sold in the whole week?

Answer: ________________

4. A fair coin is flipped and a fair number cube labelled one to six is rolled. What is the probability of getting heads *and* a five? Write it as a fraction in lowest terms.

Answer: ________________

5. Look again at the two teams' scores from the first two problems.

(a) The range of the Falcons' scores is _________.

(b) The range of the Herons' scores is _________.

(c) Which team's scoring was more spread out? Explain what that says about how consistent the team was.

6. Two fair number cubes, each labelled one to six, are rolled together. What is the probability that the two numbers add up to seven? Write it as a fraction in lowest terms.

Answer: __________________

7. Using Display A:

(a) The busiest day sold _________ sandwiches.

(b) The quietest day sold _________ sandwiches.

(c) The difference between them is _________ sandwiches.

8. A shop recorded the number of umbrellas sold on five days: 3, 5, 6, 8, 20.

(a) The median is _________.

(b) The mean is _________.

(c) Which of the two better represents a typical day? Explain your choice.

9. A bag holds 3 red marbles and 5 blue marbles. One marble is drawn, its colour recorded, and it is *put back*; then a second marble is drawn. What is the probability that both draws are red? Write it as a fraction in lowest terms.

Answer: ________________

10. Using Display A:

(a) Wednesday sold ________ sandwiches.

(b) What fraction of the week's sandwiches were sold on Wednesday? Give the fraction in lowest terms. Answer: ________

11. A spinner has four equal sections numbered one to four. It is spun twice and the two numbers are added. Use a table of all sixteen outcomes to find the probability that the total is greater than six. Write it as a fraction in lowest terms.

Answer: ________________

12. **Challenge.** Two classes sat the same quiz, marked out of twelve.

Class	**Scores**
Class One	6, 7, 7, 8, 12
Class Two	7, 8, 8, 8, 9

(a) Mean of Class One: ________ (b) Mean of Class Two: ________

(c) Range of Class One: ________ (d) Range of Class Two: ________

(e) A teacher wants the class whose results were steadier. Explain which class that is and how your four numbers decide it.